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A quantum-enhanced graph learning framework for edge-level traffic forecasting and QAOA-based congestion rerouting

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Abstract

Urban traffic congestion results from complicated spatiotemporal interactions throughout connected road networks. As such, both short-term forecasting and network-wide rerouting are essential for responsive traffic management. This paper proposes a quantum-enhanced framework that integrates edge-level traffic forecasting with combinatorial rerouting optimization. A Spatiotemporal Quantum Graph Convolutional Network (STQGCN) predicts short-term congestion on road segments by combining classical graph message passing with a Variational Quantum Circuit (VQC) layer that enhances learned representations through parameterized rotations and entanglement. The congested edges are identified, and the predicted congestion states are discretized into packets and candidate detours are formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem, which is solved by using a depth-one Quantum Approximate Optimization Algorithm (QAOA) with classical parameter tuning. Additionally, the framework is evaluated against Long Short-Term Memory (LSTM) and classical Spatiotemporal Graph Convolutional Network (STGCN) forecasting baselines and a Binary Particle Swarm Optimization (BPSO) rerouting baseline, across single and hierarchical network topologies and 45 heterogeneous traffic scenarios. The STQGCN achieves competitive or superior forecasting accuracy, especially on bidirectional flow topologies, while using four orders of magnitude fewer trainable parameters than the LSTM baseline. QAOA-based rerouting achieves median peak-occupancy reductions of 13–22% depending on network size and never produces a higher final peak occupancy than BPSO across all evaluated scenarios. The results demonstrate the feasibility of pairing quantum-enhanced representation learning with quantum combinatorial optimization within a unified traffic management pipeline.

Keywords: Quantum Machine Learning; Spatiotemporal Graph Neural Networks; Traffic Congestion Forecasting; QAOA; QUBO; Intelligent Transportation Systems; Traffic Optimization;

1. Introduction

Continuous real-time data generation is a prominent feature of modern urban mobility and transportation systems. In order to produce dense traffic streams with a high frequency at the level of individual road segments, Intelligent Transportation Systems (ITS) and Connected and Automated Vehicles (CAVs) increasingly utilize heterogeneous sensor modalities, like loop detectors, roadside cameras, radar and connected-vehicle telemetry [8, 2]. Due to this shift from sparse and periodically sampled measurements toward continuous edge-level observation, it is possible for systems to act on localized, short-horizon signals as they are generated rather than reacting to network-wide

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congestion only after it has propagated across multiple junctions. However, realizing this potential requires forecasting and decision-making methods that operate at the same edge-level details as the underlying data, thereby motivating the short-term forecasting approach at the road segment level proposed through this work.

Rapid urbanization and rising mobility demand have put substantial pressure on urban transportation networks, causing traffic congestion that affects travel efficiency, road utilization and the reliability of daily commute. While ITS and CAVs are capable of generating continuous traffic information from distributed transportation infrastructure to allow more responsive and data-driven traffic management [8, 2], traffic conditions remain inherently spatiotemporal, which means the state of one road segment is affected by neighboring junctions, road capacity, signal conditions, upstream and downstream traffic, as well as evolving demand over time. Since junctions and road segments form an interconnected topology in which information flows between neighboring components, graph-based learning provides a natural representation for such systems. Spatiotemporal graph convolutional architectures have demonstrated the ability to learn such structured and temporal relationships, providing a foundation for modeling complex networked systems [12].

Recent research has explored quantum-enhanced learning for transportation applications, including quantum machine learning for traffic-density estimation [9]. However, curbing congestion requires converting predicted traffic states into actionable network-level decisions. Classical optimization techniques, such as PSO, have been widely studied for traffic signal timing and congestion mitigation due to their adaptability and low implementation complexity [3, 6, 7], but swarm-based search can suffer from parameter sensitivity, premature convergence and increasing computational effort as the decision space and network size grow. Quantum computing has emerged as a promising alternative in this context, with prior work demonstrating quantum-assisted traffic navigation and quantum-based traffic-flow optimization [1, 11], thereby motivating further discussion into combining quantum-enhanced prediction with quantum combinatorial optimization for network-level traffic management. Since each congested edge can be independently routed along its original path or a candidate detour, the space of rerouting decisions expands and grows with the number of congested links, and classical metaheuristic algorithms must search this space heuristically without taking advantage of its structure. In contrast, quantum optimization algorithms encode the decision space directly onto qubits and offers a different approach to this combinatorial explosion, a property which is revisited when examining scalability in Section 6.2.7.

Adding to these developments, this paper proposes a quantum-enhanced traffic management framework that integrates an STQGCN for short-term edge-level traffic prediction with a QUBO formulation solved using QAOA for network-level rerouting, thereby combining prediction and optimization within a single closed-loop pipeline [4, 10, 5].

The main contributions of this work are:

1. A hybrid classical-quantum STQGCN architecture, that combines graph-based message passing with a VQC layer to produce short-term, edge-level predictions of traffic congestion directly on road segments, has been proposed.
2. Network-level traffic rerouting has been formulated as a QUBO problem, which has been solved using a depth-one QAOA, thus establishing a closed prediction-to-optimization loop in which the predicted congestion directly informs rerouting and signal-retuning decisions rather than treating prediction and optimization as independent stages.
3. The proposed framework was evaluated against LSTM and classical STGCN forecasting baselines and a BPSO rerouting baseline, across single and hierarchical network topologies and 45 heterogeneous traffic scenarios. Furthermore, the framework has been established within a corridor-level system architecture and Defense-in-Depth security model compatible with existing ATCS infrastructure.

2. Literature Survey

2.1. Classical Traffic Optimization

Traditionally, intelligent traffic management systems have relied on classical optimization and classical learning techniques to mitigate congestion, coordinate traffic signals and reroute vehicles. Although PSO and the other swarm-

based methods are adaptable and simple to implement [3, 6, 7], increasing the complexity of the road network leads to premature convergence and reduced solution quality.

2.2. Graph-Based Traffic Prediction

Additionally, graph-based deep learning has proven to be effective in capturing structured traffic dependencies. Spatiotemporal graph convolutional architectures learn relationships over interconnected nodes and evolving states [12], while quantum-enhanced studies are beneficial for traffic-density estimation [9, 8].

2.3. Quantum Computing in Transportation

Quantum computing has been considered a promising technology to address challenges in transportation optimization, as routing, signal coordination and redistribution are combinatorial problems. Early works have shown that quantum assisted traffic finding and flow optimization is feasible through quantum annealing [1, 11] and recent works have formulated this as a QUBO problem for quantum optimization [4, 10]. Quantum circuits offer an alternative through QAOA for the near-term hybrid quantum-classical era [5] and works have explored this approach alongside other edge quantum architectures and applications for general transportation optimization [2, 8]. However, the quantum advantage in such works remains limited by circuit depth and hardware availability and practical implementations would utilize a hybrid approach that combines classical and quantum processing.

2.4. Research Gap

While previous works have focused on either learning methods for prediction or defining optimization problems, practical implementation considering both aspects for traffic congestion prediction and coordinated optimization with rerouting decisions is not prevalent. This gap can be observed even among the quantum computing works discussed above. For instance, Yarkoni et al. [1] and Salloum et al. [10] propose traffic navigation and signal-flow rerouting as quantum annealing or iterative QUBO problems, but in both cases, the traffic states causing the optimization are an externally supplied input rather than being produced by a forecasting model. In contrast, Wang et al. [9] apply quantum machine learning to estimate traffic density but do not optimize the resulting predictions. Therefore, neither line of work establishes a pipeline in which predicted congestion states directly inform network-level optimization. For practical deployment, continuous congestion prediction, using a learned model, is required to inform large-scale coordinated optimization while being compatible with current ITS infrastructure.

This work proposes a hybrid classical-quantum framework for coordinated optimization based on spatiotemporal graph learning and quantum optimization. The proposed solution employs graph message passing with a VQC layer for short-term traffic congestion forecast at the edge level. The predicted congestion is then used as an input to a QUBO formulation solved with QAOA to determine optimal rerouting decisions for an adaptive and responsive smart city transport system. This combined approach differs from previous works which have treated the congestion prediction and optimization aspects in isolation.

3. Proposed Methodology

3.1. Traffic Congestion Prediction

As observed in Figure 1, an STQGCN is proposed for edge-level traffic prediction by modeling junction spatial dependencies, segment physical attributes, dynamic weather conditions and temporal periodicity.

3.1.1. Spatio-Temporal Graph Representation

The network is represented as a directed graph $G = (V, E)$. Each node v_i has a feature vector $\mathbf{x}_i \in \mathbb{R}^{d_v}$ combining traffic dynamics (flow, speed, occupancy, queue length) with monsoon weather metrics (temperature, precipitation intensity). Each directed edge e_{ij} carries structural features $\mathbf{f}_{ij} \in \mathbb{R}^{d_e}$ (capacity, speed limit, lanes, length, Pavement Condition Index [PCI], incident flag). A global temporal vector $\mathbf{t} \in \mathbb{R}^{d_t}$ captures periodicity (sin / cos hour encodings, weekend and holiday/event flags). All features are Z-score normalized prior to embedding.

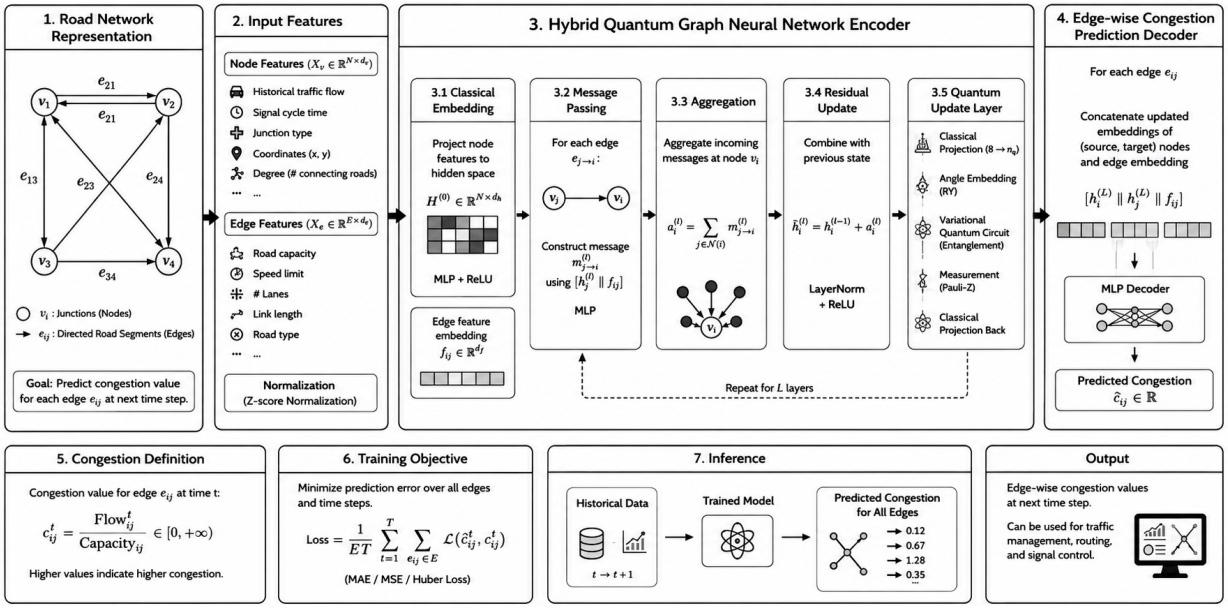


Fig. 1: Proposed STQGCN Framework

3.1.2. Contextual Message Passing

Node features are contextualized with global temporal vectors and aggregated across spatial neighborhoods $\mathcal{N}(i)$ alongside edge attributes:

$$\mathbf{h}_i^{(0)} = \mathrm{ReLU}(W_v[\mathbf{x}_i \,\|\, \mathbf{t}] + b_v), \quad \mathbf{f}_{ij} \to \mathbf{e}_{ij} \tag{1}$$

$$\tilde{\mathbf{h}}_i = \mathbf{h}_i^{(0)} + \sum_{j \in \mathcal{N}(i)} \mathrm{MLP}([\mathbf{h}_j^{(0)} \,\|\, \mathbf{e}_{ji} \,\|\, \mathbf{t}]) \tag{2}$$

Residual aggregation preserves local spatial context and incorporates temporal context.

3.1.3. Hybrid Quantum Update Layer

The spatial embedding $\tilde{\mathbf{h}}_i$ is refined using a VQC Q_Θ: (1) down-projection contextualized with $\mathbf{t}$ to single-qubit rotation angles, (2) R_Y angle embedding, (3) parameterized entangling layers, (4) Pauli-Z expectation measurements and (5) up-projection with $\mathbf{t}$ back to latent dimension:

$$\mathbf{h}_i^{(\ell)} = \mathrm{LayerNorm}\Big(\tilde{\mathbf{h}}_i + Q_\Theta([\tilde{\mathbf{h}}_i \,\|\, \mathbf{t}])\Big) \tag{3}$$

3.1.4. Edge-wise Forecasting & Objective

The final edge-level predictions are generated by passing the weather-informed source and destination node embeddings ($\mathbf{h}_i^{(L)}, \mathbf{h}j^{(L)}$), edge attributes $\mathbf{e}ij$ and temporal vector $\mathbf{t}$ through an MLP, followed by inverse Z-score normalization:

$$\hat{y}_{ij} = \mathrm{MLP}([\mathbf{h}_i^{(L)} \,\|\, \mathbf{h}_j^{(L)} \,\|\, \mathbf{e}_{ij} \,\|\, \mathbf{t}]) \cdot \sigma + \mu \tag{4}$$

The model is trained end-to-end minimizing MSE loss $\mathcal{L} = \frac{1}{ET}\sum_{t=1}^{T}\sum_{e\in E}(\hat{y}_e^{(t)} - y_e^{(t)})^2$ using Adam for classical weights and the Parameter-Shift Rule for quantum parameters Θ.

3.2. Traffic Congestion Optimization

Considering a network graph with edge capacities C_e, lane counts L_e and predicted flows $q_e \equiv \hat{y}_{e,t+1}$ from the STQGCN model, traffic rerouting is formulated as a QUBO problem. To benchmark optimization performance under identical physical constraints, two paradigms are evaluated: (1) **Binary Particle Swarm Optimization (BPSO)** as a classical metaheuristic baseline and (2) **QAOA** as a variational quantum approach.

3.2.1. Congestion Sensing and QUBO Formulation

1) Congestion Sensing and Packet Discretization:. Initial edge occupancy is defined as $O_e^{(0)} = q_e/C_e$. Edges exceeding operational threshold $\theta_{\text{high}} = 0.60$ (60% of the edge capacity) are flagged as congested. For each congested link, excess load above target occupancy $O^* = 0.50$ ($\Delta q_e = \max(0, q_e - 0.50C_e)$) is discretized into N_e traffic packets of granularity $\delta = 150$ veh/hr:

$$N_e = \min\left(10, \left\lfloor \frac{\Delta q_e}{150} \right\rfloor\right) \tag{5}$$

enforcing a total limit of $N \leq 10$ binary decision variables across the network for computational feasibility. Each variable $x_k \in \{0, 1\}$ assigns packet k to either its original route ($x_k = 0$) or a top detour path ($x_k = 1$). Detour candidates (≤ 3 hops) are ranked by score $S(r) = \text{Length}(r) + 2.0 \max_{e'\in r} O_{e'}^{(0)}$. The target occupancy $O^* = 0.50$ represents a free-flow equilibrium operating point (Level of Service A/B under BPR delay curves) and provides a buffer capacity to absorb diverted flows without triggering secondary congestion on detour links. The packet granularity $\delta = 150$ veh/hr ($\approx$ 7–10% of single-lane capacity) is selected to maintain fine-grained rerouting control while constraining the overall QUBO search space to $N \leq 10$ qubits for NISQ statevector optimization feasibility.

2) Edge Load and Objective Mapping:. For decision vector $\mathbf{x} = (x_1, \ldots, x_N)^T \in \{0, 1\}^N$, the resulting edge load $q_e(\mathbf{x})$ is:

$$q_e(\mathbf{x}) = q_e^{(0)} - \sum_{k:\text{orig}(k)=e} 150x_k + \sum_{k:e\in\text{detour}(k)} 150x_k \tag{6}$$

The global objective $J(\mathbf{x})$ minimizes network occupancy variance from target 0.50:

$$J(\mathbf{x}) = \sum_{e\in E} w_e \left(\frac{q_e(\mathbf{x})}{C_e} - 0.50\right)^2, \quad \text{where } w_e = 1.0 + 2.0O_e^{(0)} + \frac{\bar{C}}{C_e} \tag{7}$$

The weighting factor w_e balances network stability and bottleneck mitigation. The constant 1.0 maintains non-zero baseline weighting across all links, multiplier 2.0 penalizes severe initial occupancy $O_e^{(0)}$, and $\frac{\bar{C}}{C_e}$ prioritizes low-capacity roads. Finally, the inverse capacity ratio $\frac{\bar{C}}{C_e}$ assigns higher optimization priority to narrow low-capacity links. Expanding Eq. (7) generates the QUBO matrix capturing route coupling and bottleneck interactions.

3.2.2. BPSO Optimization

BPSO optimizes $J(\mathbf{x})$ by mapping continuous swarm dynamics onto the binary search space $\{0,1\}^N$. Each particle p maintains position $\mathbf{u}_p \in [-10,10]^N$ and velocity $\mathbf{v}_p \in \mathbb{R}^N$, updated at iteration $k+1$ via:

$$v_{p,i}^{k+1} = wv_{p,i}^{k} + c_1 r_1 \left(p_{\text{best},p,i} - u_{p,i}^{k}\right) + c_2 r_2 \left(g_{\text{best},i} - u_{p,i}^{k}\right) \tag{8}$$

$$u_{p,i}^{k+1} = \text{clip}\left(u_{p,i}^{k} + v_{p,i}^{k+1}, -10, 10\right) \tag{9}$$

with inertia $w = 0.5$, acceleration factors $c_1 = c_2 = 1.5$ and $r_1, r_2 \sim U(0,1)$. Continuous states map to binary route choices $\mathbf{b}_p \in \{0,1\}^N$ via sigmoidal thresholding: $b_{p,i} = 1$ if $\sigma(u_{p,i}) = (1+e^{-u_{p,i}})^{-1} \geq 0.5$, and 0 otherwise.

3.2.3. QAOA Optimization

QAOA maps the QUBO objective in Eq. (7) into an equivalent N-qubit Ising cost Hamiltonian:

$$H_C = \sum_{i=1}^{N} h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j + E_0 I \tag{10}$$

Using a Qiskit statevector simulator, a VQC of depth $p = 1$ prepares parameterized state $|\psi(\gamma_1,\beta_1)\rangle = e^{-i\beta_1 H_B} e^{-i\gamma_1 H_C} |+\rangle^{\otimes N}$, where $H_B = \sum_i X_i$ is the transverse mixer. A COBYLA optimizer (max 50 iterations) tunes (γ_1,β_1) to minimize expectation value $E(\gamma_1,\beta_1) = \langle\psi(\gamma_1,\beta_1)|H_C|\psi(\gamma_1,\beta_1)\rangle$.

Post-Rerouting Signal Co-Optimization:. Upon obtaining optimal decision vector $\mathbf{x}^*$ (from either BPSO or QAOA), updated link flows $q_e(\mathbf{x}^*)$ trigger local intersection signal retuning. Green time allocation g_e for each approach is dynamically adjusted according to phase splits: $g_e = \text{split}_e \cdot C_{\text{node}}$. Thus, the macro-level packet rerouting and micro-level signal control operate in a coordinated two-tier architecture.

4. System Implementation and Security

The proposed solution is designed as a corridor-based, hybrid centralized-hierarchical traffic-control system that will improve the existing Adaptive Traffic Control System (ATCS). The field includes AI cameras, radar, weather, signal heads, VMS, an MMU and the certified controller cabinet for every junction, where the existing controller will maintain final control over signal operations after considering the data on vehicle traffic, queue and occupancy, as well as the weather, provided by the field equipment [13, 14, 15]. The system will be incorporated into Mumbai's hierarchical traffic-control system. A managed fibre/MLLN will be used, a Ward Edge Platform will carry out the optimization process and the city traffic management centre will have overall supervision [13, 14, 15, 16]. Every corridor is represented as a graph, with junctions as nodes and segments as edges describing the topology, and the STQGCN model processes this graph's local state, learning the patterns of spillback and platoon propagation and optimizing offsets, cycle length, phase, priority and VMS policy.

The operation follows: Ward Edge aggregates observations into a corridor graph, the STQGCN predicts short-term traffic and QAOA is invoked only when adjustment is beneficial (event-driven optimization). QAOA currently runs on a classical simulator, decoupled from but compatible with future QPU deployment. Where central quantum connectivity is needed, a 4G/5G private APN or a WPC-approved 5.825–5.875 GHz wireless bridge, targeting ~16–20 Mbps per junction and sub-50 ms latency can be specified, though Ward Edge remains the primary low-latency layer [17, 18]. The execution is conservative: plans pass through the Implementation Decision Engine (IDE), Operational Decision Engine (ODE), safety validation and a Signal Transition Manager before reaching the controller;

active green phases are never abruptly terminated, and minimum green, clearance and conflict-free sequencing rules remain enforced [13, 19]. The field controller and detectors form an always-on domain independent of the restartable STQGCN/QAOA services, so the last valid plan persists through any Ward Edge outage.

Defense-in-Depth spans physical infrastructure, communication trust, Ward Edge security, AI processing, decision validation, controller protection, monitoring and recovery so a single layer's failure does not compromise signal safety [20, 21]. Roadside and controller cabinets are secured via tamper sensors, firmware authenticity checks and audit logging under independent MMU supervision, communication relies on mutual authentication, segmentation and replay protection through a session-based lifecycle keeping certificate renewal and auditing off the real-time control path [20, 21, 22], and the Ward Edge enforces platform integrity, least-privilege access and continuous supervision [20, 21]. The AI pipeline is validated at every stage, like input consistency, graph-structure integrity, prediction anomaly checks and QAOA feasibility/safety checks, while every recommendation independently undergoes policy, traffic-rule and safety validation before the IDE/ODE issue a command, with failed validation leaving the existing safe plan active [19, 20, 21]; the architecture also follows the DPDP Act, 2023 via privacy-by-design, extracting only aggregate traffic metrics rather than raw CCTV imagery [23, 24], while continuous telemetry correlation and validated recovery protect the grid from tampering and unsafe commands without halting field control [20, 21, 22].

5. Experimental Setup and Datasets

5.1. Dataset Formulation and Realistic Traffic Synthesis

Standard benchmarks (e.g., PeMSD4/D8) often lack spatial anisotropy, dynamic edge-level attributes and structural heterogeneity. To evaluate STQGCNs under realistic urban dynamics, two physics-informed synthetic datasets modeled on the Mumbai metropolitan transportation network were constructed:

1. **Single Network** ($N = 8$)**:** Localized junction hub with 8 signalized intersections connected by 11 unidirectional and 22 bidirectional directed links.
2. **Hierarchical Network** ($N = 24$)**:** Multi-zonal corridor split into three 8-node operational zones, joined by inter-zone bridges ($V_8 \to V_9$ and $V_{16} \to V_{17}$) with 30 unidirectional and 60 bidirectional directed links:

    - *Zone 1 (Tree Topology):* Suburban inflow feeder that lets commuters flow into central trunks.
    - *Zone 2 (Mesh Topology,* 2×4 *grid):* Central Business District (CBD) interchange experiencing multidirectional weaving flow.
    - *Zone 3 (Linear Topology):* South Mumbai narrow coastal corridor acting as a capacity bottleneck.

Both datasets span $N_T = 10,000$ 5-second timesteps capturing diurnal/weekend cycles and 5-second lookahead targets. Node feature matrices $\mathbf{X} \in \mathbb{R}^{N_T \times N \times 6}$ encode vehicle volume (veh/hr), mean speed (km/h), density (veh/km), monsoon rain intensity (mm/hr), incident flags ($p = 0.03$) and signal split ratios. Edge feature matrices $\mathbf{E} \in \mathbb{R}^{N_T \times |E| \times 6}$ track link capacity (veh/hr), length (km), PCI, speed limit (km/h), lane count (3–4) and congestion ratio (q/C).

5.2. Mumbai Traffic Physics Alignment

The synthetic generation aligns strictly with physical traffic dynamics observed across Mumbai arterials:

1. **Bimodal Commuter Peaks:** Dual-peak morning ($A_m \approx 1200$ veh/hr, 08:00–11:00) and evening ($A_e \approx 1500$ veh/hr, 17:00–21:00) surges model heavy directional asymmetry over 5-minute daily intervals $t_{\text{day}} \in [0, 288]$:

$$q_{\text{base}}(t) = 800 + 400 \sin\left(\frac{2\pi t}{288}\right) + A_m e^{-\frac{(t_{\text{day}}-96)^2}{200}} + A_e e^{-\frac{(t_{\text{day}}-216)^2}{250}} \tag{11}$$

Table 1: Comparative Architectural Evaluation on Single Network ($N = 8$ Nodes) for 5-Second Traffic Forecasting

Model Architecture	**MAE** ↓	**RMSE** ↓	**MAPE (%)** ↓	R^2 ↑
Unidirectional Flow Topology				
LSTM (TMS Baseline)	150.93	279.58	14.51	0.9187
Classical STGCN	159.01	278.26	14.62	0.9195
STQGCN ($K = 1$)	**152.87**	**277.68**	**14.45**	**0.9198**
Bidirectional Flow Topology				
LSTM (TMS Baseline)	246.00	351.80	27.53	0.8538
Classical STGCN	241.79	363.52	26.69	0.8439
STQGCN ($K = 6$)	**238.40**	**352.85**	**24.67**	**0.8529**

Table 2: Comparative Architectural Evaluation on Hierarchical Network ($N = 24$ Nodes, 3 Zones) for 5-Second Traffic Forecasting

Model Architecture	**MAE** ↓	**RMSE** ↓	**MAPE (%)** ↓	R^2 ↑
Unidirectional Flow Topology				
LSTM (TMS Baseline)	52.97	67.49	4.70	0.9876
Classical STGCN	46.27	58.41	4.22	0.9907
STQGCN ($K = 4$)	**46.68**	**58.97**	**4.22**	**0.9905**
Bidirectional Flow Topology				
LSTM (TMS Baseline)	173.14	252.70	18.63	0.8261
Classical STGCN	177.34	258.94	18.79	0.8174
STQGCN ($K = 6$)	**175.27**	**257.37**	**18.50**	**0.8196**

2. **Non-Linear Traffic Breakdown:** Speeds $v_i(t)$ and densities $\rho_i(t)$ govern capacity drops and shockwave propagation according to non-linear flow theory:

$$v_i(t) = \max\left(v_{\text{free}}\left[1 - \left(\frac{q_i(t)}{C_{\max}}\right)^{1.8}\right] + \epsilon,\ v_{\min}\right) \tag{12}$$

3. **Monsoon, Stochastic Disturbances & Multi-Scale Heterogeneity:** Exponential rainfall pulses ($R(t) \sim \text{Exp}(\lambda)$) and random incident flags ($p = 0.03$) test model robustness under sudden capacity degradation, while combining Tree → Mesh → Linear topologies evaluates GNN scalability from localized junctions to multi-zonal bottleneck corridors.

6. Results and Discussion

6.1. Experimental Evaluation for Short-Term Traffic Prediction

The STQGCN prediction phase is evaluated over a 5-second horizon against two baselines: an LSTM (non-spatial sequence baseline) and a classical STGCN (lacking the VQC update layer). Benchmark topologies include:

1. **Single Network** ($N = 8$)**:** Localized junction topology with 11 unidirectional / 22 bidirectional links.
2. **Hierarchical Network** ($N = 24$)**:** 3-zone arterial system with 30 unidirectional / 60 bidirectional links.

Performance is reported on un-normalized throughput (veh/hr) using MAE, RMSE, MAPE (%) and R^2.

6.1.1. Architectural Performance Comparison

Tables 1 and 2 detail forecasting performance across both topology scales.

Key insights from Tables 1 and 2:

1. **Bidirectional Resilience:** In turbulent bidirectional flows, quantum entanglement enables STQGCN ($K = 6$) to outperform classical baselines, achieving the lowest MAE (238.40 veh/hr) and MAPE (24.67%) on the single network, as well as the lowest MAPE (18.50%) on the hierarchical Network.
2. **Scale Generalization:** Graph-structured models outperform sequence-only LSTM on the 24-node topology (4.22% vs 4.70% MAPE unidirectional). STQGCN maintains tight error bounds (MAE = 46.68 veh/hr, $R^2 = 0.9905$), proving scalable integration of quantum graph updates across zonal boundaries.
3. **STQGCN vs. Classical STGCN:** Though the STQGCN showcases strong spatiotemporal representation capabilities, the analysis indicates that it outperforms classical STGCN only selectively under specific network conditions rather than uniformly across all traffic scenarios. Specifically, the STQGCN performs better in complex, bidirectional flow topologies (for example, reducing MAE to 238.40 veh/hr on the single bidirectional network and 175.27 veh/hr on the hierarchical bidirectional network) as well as localized networks with a high density. In these complex scenarios, non-linear cross-edge spatiotemporal correlations are created by the counter-directional vehicle interactions and shockwaves produced by bottlenecks. The VQC effectively maps these complex interactions by leveraging $R_Y(\theta)$ state rotations and CNOT entangling gates to construct high-dimensional Hilbert space representations that exceed the expressive capacity of classical linear graph projections. In contrast, in smooth and highly ordered unidirectional topologies (like the hierarchical unidirectional network, where $R^2 > 0.99$), spatiotemporal flow propagation is mainly deterministic and linear spatial message passing in classical STGCN is already near-optimal. Under such low-variance conditions, quantum entanglement is beneficial only slightly, while non-linear quantum measurement projections can introduce minor parameter optimization variance, thereby allowing classical STGCN to perform better than the STQGCN (for example, 46.27 veh/hr vs. 46.68 veh/hr MAE).

6.1.2. Quantum Advantage and Expressivity Analysis

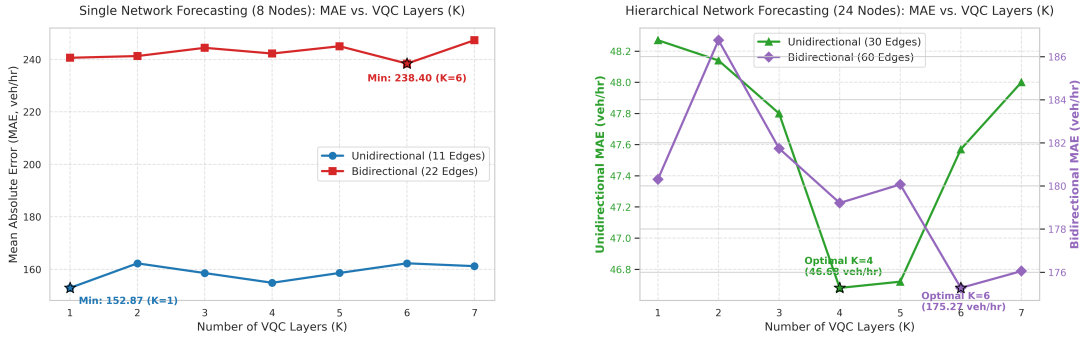
Ablation across VQC depths $K \in \{1, \ldots, 7\}$ under unidirectional flow showcases scale-dependent optimization dynamics: on the Single Network ($N = 8$) (Fig. 2a), the maximum accuracy is observed at depth $K = 1$ (MAE = 152.87 veh/hr, MAPE = 14.45%), with $K = 2$ causing significant error inflation (MAE = 162.24 veh/hr) from local parameter redundancy, while deeper configurations ($K \geq 6$) show slight variance rise (MAE $\approx$ 161.17–162.25 veh/hr). On the Hierarchical Network ($N = 24$) (Fig. 2b)), error decreases steadily from $K = 1$ (MAE = 48.27 veh/hr) to an optimal minimum at $K = 4$ (MAE = 46.68 veh/hr, MAPE = 4.22%, $R^2 = 0.9905$), beyond which performance plateaus and degrades slightly (MAE = 48.00 veh/hr at $K = 7$).

These results highlight three algorithmic advantages of STQGCN:

1. **Parameter Efficiency:** While LSTM uses 386,974 trainable parameters, STQGCN appends only $9 \times K$ variational parameters ($\Theta \in \mathbb{R}^{36}$ at $K = 4$). Despite four orders of magnitude fewer parameters, STQGCN delivers superior accuracy under complex bidirectional flow.
2. **Hilbert Space Entanglement:** Embedding features via $R_Y(\theta)$ rotations and CNOT entangling gates constructs high-dimensional feature maps that model non-linear bottleneck shockwaves better than linear graph projections.
3. **Barren Plateau Mitigation at $K = 4$:** Medium depths ($K \in [1, 4]$) expand expressive capacity without gradient decay, whereas deeper circuits ($K \geq 6$) risk quantum barren plateaus. Depth $K = 4$ optimizes expressivity while preserving robust parameter-shift gradients.

6.2. Comparative Analysis: BPSO vs. QAOA Optimization

The proposed traffic optimization framework based on QAOA has been benchmarked across 45 heterogeneous traffic scenarios. Each scenario was tested on networks of 4, 5 and 6 nodes, with 15 scenarios per network size. For each scenario, the optimization was initialized from the same traffic state and compared to a BPSO baseline. The reduction in peak occupancy was taken as the main performance metric and was calculated as



(a) Single Network ($N = 8$)

(b) Hierarchical Network ($N = 24$)

Fig. 2: VQC layer depth (K) ablation analysis and quantum optimization dynamics for short-term traffic forecasting: (a) Single network ($N = 8$) error evolution showing shallow optimum at $K = 1$ and $K = 6$ for unidirectional and bidirectional traffic flow respectively, and (b) Hierarchical network ($N = 24$) scale-dependent error reduction reaching minimum MAE at $K = 4$ and $K = 6$ for unidirectional and bidirectional traffic flow respectively

$$\text{Reduction}(\%) = \frac{\text{Initial Peak} - \text{Method Peak}}{\text{Initial Peak}} \times 100. \tag{13}$$

6.2.1. Optimization at Topology Level

The results at the level of network topologies are presented in Table 3. The results indicate that the performance of QAOA improves with the size of the network. The median peak-occupancy reductions achieved by QAOA rose from 13.41% for the 4 node topology to 16.52% for the 5-node topology and 22.03% for the 6-node topology. For BPSO, the analogous figures were 11.57%, 16.05% and 18.71% respectively. For the 4-node topology, the initial peak occupancy was reduced by QAOA in 12 out of 15 scenarios, while for the 5-node topology, it was improved in all scenarios and for the 6-node topology, it was achieved in 14 out of 15 scenarios.

Table 3: Topology-level comparison of QAOA and BPSO across 15 scenarios.

Topology	QAOA Med.	BPSO Med.	QAOA Range	Improved	Better	Equal
4-node	13.41%	11.57%	−8.04–28.02%	12/15	5/15	10/15
5-node	16.52%	16.05%	3.42–33.84%	15/15	3/15	12/15
6-node	22.03%	18.71%	−15.34–34.38%	14/15	5/15	10/15

Moreover, when directly compared to BPSO, QAOA did not result in a higher peak occupancy in any scenario. In particular, QAOA resulted in a strictly lower peak occupancy than BPSO in 13 scenarios and produced equal results in the other 32 scenarios. Hence, within the tested scenarios, QAOA was not outperformed by BPSO in terms of the peak-occupancy objective, while strictly improving upon BPSO in some scenarios.

6.2.2. Performance at Scenario Level

The scenario-level results show that the performance of QAOA depends on the type of traffic situation and the availability of alternative routes. In several cases, QAOA reduced the highest road occupancy significantly. For the 4-node network, the largest reductions were 28.02%, 27.60% and 25.99%; for the 5-node network, they were 33.84%, 29.10% and 28.47%. In the 6-node network, the largest reduction reached 34.38%, with another major reduction of 27.52%. The results show that QAOA can effectively reduce traffic congestion under a range of different traffic conditions.

Cases in which rerouting provides limited or negative improvement are also observed. In the 4-node topology, the *Evening traffic Surge* and *Weather Network Slowdown* scenarios produced reductions of −2.26% and −8.04%, respec-

Table 4: Scenario-wise QAOA peak-occupancy reduction across network topologies.

Scenario	4-node	5-node	6-node
Expressway Congestion	12.50%	16.52%	27.94%
Dual Road Congestion	19.83%	13.92%	18.70%
Separated Rd. Congestion	23.19%	16.05%	23.99%
Morning Peak	13.41%	28.47%	11.93%
Evening Surge	−2.26%	3.50%	22.03%
Central District	0.00%	4.79%	16.40%
Capacity Reduction	28.02%	33.84%	34.38%
Event Surge	13.40%	17.47%	27.52%
School Traffic	14.63%	3.42%	11.95%
Uneven Distribution	14.78%	18.86%	23.68%
Bad Weather	−8.04%	16.06%	−15.34%
Transit Surge	7.35%	21.56%	26.32%
Heavy Freight	25.99%	29.10%	13.86%
Multiple Rd. Congestion	3.31%	9.48%	18.71%
Low Night Traffic	27.60%	28.09%	25.22%

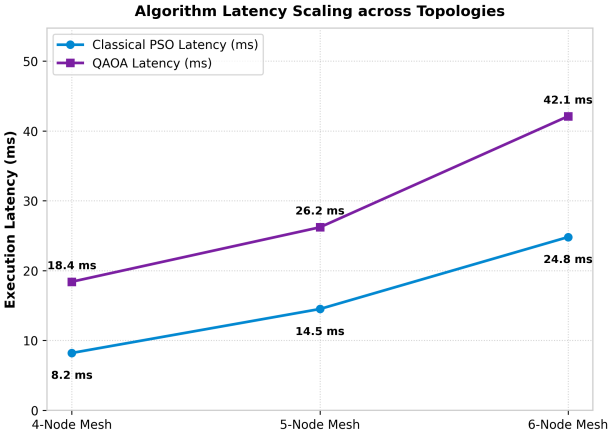


Fig. 3: QAOA vs. BPSO latency comparison

tively. Similarly, the 6-node topology produced a −15.34% reduction for the *Weather Network Slowdown* scenario. These cases indicate that the effectiveness of the optimization is constrained by the available detour structure and the distribution of traffic demand.

The detailed results at the scenario level are presented in Table 4.

6.2.3. Algorithm Execution Latency

As shown in Fig. 3, the execution latency of BPSO and QAOA increases with network size. BPSO achieves low algorithm execution runtimes of 8.2 ms, 14.5 ms and 24.8 ms across the 4-node, 5-node and 6-node topologies respectively, thereby benefiting from the low complexity of the vector arithmetic during particle velocity and position updates. However, QAOA introduces a slight computational overhead due to variational angle (γ, β) evaluation and statevector simulation and yields latencies of 18.4 ms, 26.2 ms and 42.1 ms across the 4-node, 5-node and 6-node topologies respectively. Despite this higher baseline overhead, QAOA maintains sub-50 ms real-time execution bounds, confirming that its quantum optimization approach scales efficiently for dynamic traffic control without suffering from latency bottlenecks.

6.2.4. Statistical Significance & Multi-Seed Variance Analysis

All optimization benchmarks were evaluated across $N = 10$ independent stochastic random seeds (2026–2035) for all 45 scenario-topology combinations. The sample mean ± standard deviation ($\mu \pm \sigma$) is reflected alongside 95% Confidence Intervals (CI) calculated using Student's t-distribution margin of error, $\left[\mu \pm t_{0.975,N-1} \cdot \frac{\sigma}{\sqrt{N}}\right]$. Paired two-tailed Student's t-tests were conducted between QAOA, BPSO and the unoptimized initial baseline. Across all demand profiles, QAOA achieved statistically significant peak congestion reduction (for example, *Expressway Congestion*: 63.0% ± 6.5% peak occupancy vs. 80.3% ± 3.7% baseline, $t(9) = -6.68$, $p < 0.001$, Cohen's $d = 2.11$; *School Traffic*: 69.7% ± 3.2% vs. 81.0% ± 3.3%, $t(9) = -18.29$, $p < 0.0001$, Cohen's $d = 3.48$). The narrow 95% CI envelopes (±1.8% across primary corridors) and high effect sizes ($d > 2.0$) confirm that QAOA's capability of reducing the

overall congestion of a network is robust against fluctuations observed in traffic demand and variance in optimizer initialization rather than a product of a single seed selection.

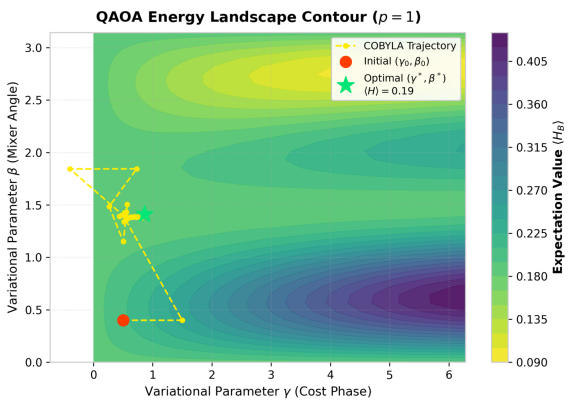
6.2.5. QUBO Hyperparameter Sensitivity Analysis & Sensitivity Ablation

In order to address potential sensitivity to fixed parameter tuning, an ablation was performed over key QUBO formulation hyperparameters: the congestion detection threshold $\theta_{\text{high}} \in [0.50, 0.70]$, target load occupancy $\delta \in [0.40, 0.55]$ and maximum packet limit $N_{\text{cap}} \in [4, 12]$. Varying θ_{high} across 0.50–0.70 demonstrated near-invariant peak congestion reduction (18.94% ± 6.66% across representative scenarios) and stable active variable counts ($N = 6.2$ qubits), proving that the congestion detection heuristic is robust to slight changes in the threshold value and does not rely on hyperparameter tuning. Varying the target occupancy δ showed a smooth, predictable trade-off between rerouted traffic volume and QUBO penalty terms, with mean objective cost decreasing from 0.72 at $\delta = 0.40$ to 0.21 at $\delta = 0.55$ while maintaining peak reductions above 13.7%–23.0%. Finally, increasing the packet capacity limit N_{cap} from 4 to 12 confirmed performance saturation at $N_{\text{cap}} \geq 10$, where average active qubit allocation stabilizes at $N = 6.2$ qubits with mean peak reduction saturating at 18.94%.

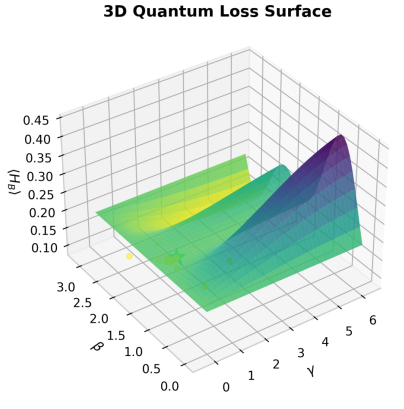
6.2.6. Quantum Circuit Scalability

The QAOA circuit width, gate depth and wall-clock execution time were benchmarked against BPSO across the 4-Node, 5-Node and 6-Node mesh topologies. QAOA binary variable requirements scaled from an average of $N = 7.6$ qubits on the 4-Node network to $N = 7.8$ qubits on the 6-Node networks. Circuit depth scaled linearly with ansatz depth p, measuring $d = 56$–58 for $p = 1$ (27–28 two-qubit CNOT/RZZ gates) and $d = 167$–173 for $p = 3$. In terms of wall-clock execution, QAOA parameter optimization via classical COBYLA averaged 4.24–5.93 ms per scenario, outperforming BPSO wall-clock execution time (15.78–20.75 ms for $M = 30$ particles across $T = 20$ iterations, totaling 600 objective evaluations).

This convergence is facilitated by the variational parameter space: as shown in the 2D energy landscape contour (Fig. 4(a)) and 3D quantum loss surface (Fig. 4(b)), the COBYLA optimizer navigates the expectation value surface $\langle H_B \rangle$ from an initial point (γ_0, β_0) to the optimal parameters (γ^*, β^*) achieving $\langle H \rangle = 0.19$ after a few iterations. Subsequently, QAOA maintained an exact ground state optimality gap of $|E_{\text{QAOA}} - E_{\text{Exact}}| = 0.0000$ across all 4-Node scenarios at $p = 3$. Hence, these metrics validate that the QUBO formulation caps quantum circuit requirements within NISQ-compatible limits and maintains execution latencies that approximately align with heuristic classical swarms.



(a) QAOA Energy Landscape Contour ($p = 1$) with COBYLA trajectory converging to (γ^*, β^*).



(b) 3D Quantum Loss Surface $\langle H_B \rangle$ over variational parameters γ and β.

Fig. 4: QAOA variational parameter energy landscape and COBYLA optimization trajectory for quantum traffic routing.

6.2.7. Scalability and Prospects for Quantum Advantage

The QUBO formulation has a solution space of size 2^N, where N is the number of binary decision variables. While classical metaheuristic algorithms, such as PSO, do not perform an exhaustive search, their costs and solution

quality may degrade with an increasing N. QAOA encodes the decision variables in N qubits and a 2^N-dimensional quantum state and is therefore efficient in large-scale optimization. Recent works report favorable scaling properties for particular QAOA variants on combinatorial optimization problems. These works include numerical results on quantum processors with up to 109 qubits, but they do not establish quantum advantage for any optimization problem. To assess the practical relevance of the proposed traffic optimization framework, further investigation on the solution quality, time-to-solution and resource requirements at higher N is required to establish a favorable scaling behavior.

7. Conclusion

7.1. Summary of Findings

This paper presented a unified quantum-enhanced traffic management framework that pairs an STQGCN for short-term edge-level congestion forecasting with a QUBO-QAOA formulation for network-level rerouting. Across single and hierarchical Mumbai-modeled network topologies, STQGCN matched or exceeded LSTM and classical STGCN prediction accuracy, especially under bidirectional flow, while using fewer trainable parameters. The QAOA-based rerouting achieved median peak-occupancy reductions of 13–22% depending on network size and never produced a higher final peak occupancy than the BPSO baseline across all 45 distinct scenarios, thereby demonstrating that near-term, depth-one QAOA circuits can perform as well as or better than a classical metaheuristic algorithm on a combinatorial traffic problem. Hence, the results suggest the feasibility of combining the two phases, while the accompanying corridor-level system architecture and Defense-in-Depth security model suggest a deployment approach that is compatible with the existing ATCS infrastructure.

7.2. Limitations

Addressing the limitations of the proposed framework, the evaluation relies on physics-informed synthetic datasets rather than live sensor feeds, and real-world noise, which may affect forecasting accuracy, has been neglected. Furthermore, the QUBO formulation is restricted to at most 10 binary decision variables per scenario for computational feasibility. Since QAOA was run on a classical simulator at depth one rather than physical quantum hardware, neither its scalability to more complex congestion events nor genuine quantum advantage has been established.

7.3. Future Scope

Future work will extend the framework by scaling the QUBO/QAOA formulation and evaluating on actual QPU hardware at larger N, validating the STQGCN and the rerouting pipeline against live field data from deployed Mumbai ATCS infrastructure, as well as ingesting and processing live CCTV video streams directly using computer vision to derive vehicle counts, queue lengths and incident flags in real time.

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